

Comprehensive Road Accident Analysis Report

Overview of Data The dataset contains detailed records of road traffic accidents totaling 307,973 incidents. These accidents resulted in 417,883 casualties and involved 563,302 vehicles. The data spans across two years, with 163,554 accidents recorded in 2021 and 144,419 accidents in 2022.

Accident Severity The recorded accidents are classified into three main severity levels:

- Slight accidents: Account for the vast majority at 85.49 percent of all incidents.
- Serious accidents: Form 13.23 percent.
- Fatal accidents: Represent 1.28 percent of the total recorded accidents.

Top Vehicle Types Involved Private cars lead the list by a significant margin compared to other modes of transport:

- Private cars: 239,794 vehicles.
- Light commercial vans (goods 3.5 tonnes or under): 15,695 vehicles.
- Motorcycles over 500cc: 11,226 vehicles.
- Buses or coaches: 8,686 vehicles.
- Motorcycles 125cc and under: 6,852 vehicles.

Environmental and Weather Conditions

- Weather conditions: The vast majority of accidents occurred in relatively clear weather without high winds, totaling 244,496 incidents, followed by accidents during rainy conditions without high winds at 34,877 incidents, and snowy conditions at 4,839 incidents.
- Road surface conditions: Dry surfaces experienced the highest share of accidents due to traffic volume, followed by wet or damp roads.

Road Types and Areas

- Road type: Single carriageway roads top the list with 230,612 accidents, followed by dual carriageways with 45,467 accidents, and roundabouts with 20,929 accidents.
- Geographical distribution: Urban areas account for the highest proportion at 64.46 percent of accidents, while rural areas account for 35.54 percent.

Temporal Distribution of Accidents

- Months: Accidents peaked noticeably during November with 29,095 incidents and October with 28,368 incidents, while February recorded the lowest with 21,885 incidents.
- Days of the week: Friday is the most frequent day for accidents with 50,529 incidents, followed by Tuesday, Wednesday, and Thursday with averages ranging between 45,000 and 46,000 incidents, while Sunday recorded the lowest count at 33,544 incidents.

Speed Limits The vast majority of accidents are concentrated in areas with a speed limit of 30 mph, totaling 200,040 incidents, followed by roads with a speed limit of 60 at 46,826 incidents, a speed limit of 40 at 25,650 incidents, and a speed limit of 70 at 22,362 incidents.

Conclusions and Proposed Recommendations

1. Intensify traffic monitoring and control within urban areas, which experience the highest proportion of accidents involving private cars.
2. Implement engineering measures and additional advisory signage on single carriageway roads due to high accident rates.
3. Pay special attention to peak accident seasons such as autumn months and weekend days like Friday to enhance security patrols and guidance.